Quiz 6

April 21, 2025

Answer all questions. Closed book, closed notes. Time: 20 min

Suppose 60 students of DAU have chosen ASTDS as an elective in 2025. The grading policy stipulates that 40% of the marks in the course will be assigned based on the best four scores from six surprise quizzes.

Suppose that quizzes Q1, Q2, Q3 happened on normal days and the quizzes Q4, Q5 and Q6 happened on days followed by a long weekend. As it happens, some students missed some quizzes. Suppose $(x_{i1}, x_{i2}, ..., x_{i6})$ denotes the marks of the i-th student $(i = 1, 2, ..., 60)$ in the quizzes $(Q_1, Q_2, ..., Q_6)$. It is observed that for students $i = 1, 2, ..., 35$ the marks of all quizzes are available. For students with serial number $i = 36, 37, ..., 40$ the marks in a quiz are missing if his or her marks in the previous quiz are below 20%, for students with serial number $i = 41, 42, ..., 50$ the marks in Q4, Q5 and Q6 are missing, and for the remaining students the scores are missing for reasons completely at random.

(i) Write the data matrix $D(60 \times 6)$ and missingness indicator matrix $M(60 \times 6)$. Clearly, in the last ten rows of D you need to choose the missing values at random.
Points: 2 + 2 = 4

Solution:

The data matrix $D(60 \times 6)$ contains quiz scores:

$$D = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{35,1} & x_{35,2} & x_{35,3} & x_{35,4} & x_{35,5} & x_{35,6} \\ x_{36,1} & x_{36,2} & x_{36,3} & x_{36,4} & x_{36,5} & x_{36,6} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{40,1} & x_{40,2} & x_{40,3} & x_{40,4} & x_{40,5} & x_{40,6} \\ x_{41,1} & x_{41,2} & x_{41,3} & M & M & M \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{50,1} & x_{50,2} & x_{50,3} & M & M & M \\ x_{51,1} & x_{51,2} & x_{51,3} & x_{51,4} & M & x_{51,6} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ M & x_{60,2} & x_{60,3} & M & M & x_{60,6} \\ x_{60,1} & x_{60,2} & x_{60,3} & x_{60,4} & x_{60,5} & x_{60,6} \end{pmatrix},$$

where x_{ij} is student i's score in quiz j. For $i = 1, \ldots, 35$, all scores are observed. For $i = 36, \ldots, 40$, quiz j score is missing (denoted X) if $x_{i,j-1} < 0.2 \times$max score. For $i = 41, \ldots, 50$, scores for $j = 4, 5, 6$ are X. For $i = 51, \ldots, 60$, some x_{ij} are randomly X.

The missingness indicator matrix $M(60 \times 6)$ has $M_{ij} = 1$ if x_{ij} is observed, 0 if missing:

$$M = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & m_{36,2} & m_{36,3} & m_{36,4} & m_{36,5} & m_{36,6} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & m_{40,2} & m_{40,3} & m_{40,4} & m_{40,5} & m_{40,6} \\ 1 & 1 & 1 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & 0 & 0 & 0 \\ m_{51,1} & m_{51,2} & m_{51,3} & m_{51,4} & m_{51,5} & m_{51,6} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ m_{60,1} & m_{60,2} & m_{60,3} & m_{60,4} & m_{60,5} & m_{60,6} \end{pmatrix},$$

where:

- Rows $1, \ldots, 35$: All 1.
- Rows $36, \ldots, 40$: $m_{ij} = 0$ if $x_{i,j-1} < 0.2 \times$ max score, else 1; $m_{i1} = 1$.
- Rows $41, \ldots, 50$: 1 for $j = 1, 2, 3$, 0 for $j = 4, 5, 6$.

- Rows $51, \ldots, 60$: Random 0 or 1.

continued

(ii) Define the missingness mechanisms with justification for the three groups of students with serial numbers $36, 37, ..., 40$; with serial numbers $41, 42, ..., 50$; and, with serial numbers $51, 52, ..., 60$.
Points: 4 + 4 + 4 = 12

Solution:

- **Students** $i = 36, \ldots, 40$: **MAR**. Missingness of quiz j depends on the observed score $x_{i,j-1} < 0.2 \times$ max score. MAR occurs when $P(M \mid D) = P(M \mid D_{\text{obs}})$, i.e., missingness depends on observed data.
- **Students** $i = 41, \ldots, 50$: **MCAR**. Quizzes Q_4, Q_5, Q_6 are missing due to timing (pre-long weekend day), not related to data values. Whether the value would be missing depends on how the date of the quizzes are selected. If these are selected at random then we can consider it as MCAR, if it is chosen based on the pattern of teaching, then it is not MCAR.
- **Students** $i = 51, \ldots, 60$: **MCAR**. Scores are missing "completely at random," implying $P(M \mid D) = P(M)$, a random process.

(iii) Show that under MAR
$P(D_{obs}, M|\theta, \gamma)$ is proportional to $\int_{D_{mis}} P(D|\theta)\, dD_{mis}$
Points: 4

Solution:
The likelihood is:

$$P(D_{\text{obs}}, M \mid \theta, \gamma) = \int_{D_{\text{mis}}} P(D \mid \theta) P(M \mid D, \gamma)\, dD_{\text{mis}}.$$

Under MAR, $P(M \mid D, \gamma) = P(M \mid D_{\text{obs}}, \gamma)$. Thus:

$$P(D_{\text{obs}}, M \mid \theta, \gamma) = \int_{D_{\text{mis}}} P(D \mid \theta) P(M \mid D_{\text{obs}}, \gamma)\, dD_{\text{mis}}.$$

Factor out $P(M \mid D_{\text{obs}}, \gamma)$:

$$P(D_{\text{obs}}, M \mid \theta, \gamma) = P(M \mid D_{\text{obs}}, \gamma) \int_{D_{\text{mis}}} P(D \mid \theta)\, dD_{\text{mis}}.$$

Since $\int_{D_{\text{mis}}} P(D \mid \theta)\, dD_{\text{mis}} = P(D_{\text{obs}} \mid \theta)$:

$$P(D_{\text{obs}}, M \mid \theta, \gamma) \propto \int_{D_{\text{mis}}} P(D \mid \theta)\, dD_{\text{mis}}.$$